
Problem 9.13. Consider the function *pad*: $\Sigma \times \mathbb{N} \longrightarrow \Sigma^*\#^*$ that is defined as follows. Let $pad(s, l) = s\#^j$, where $j = \max(0, l - m)$ and m is the length of s. Thus, $pad(s, l)$ simply adds enough copies of the new symbol # to the end of s so that the length of the result is at least l. For any language A and function f: $\mathbb{N} \longrightarrow \mathbb{N}$, define the language $pad(A, f)$ as

$$pad(A, f) = \{pad(s, f(m)) \mid \text{where } s \in A \text{ and } m \text{ is the length of } s\}.$$

Prove that if $A \in \text{TIME}(n^6)$, then $pad(A, n^2) \in \text{TIME}(n^3)$.

Proof. To show $pad(A, n^2) \in \text{TIME}(n^3)$, we give a TM M that decides $pad(A, n^2)$ in $\text{TIME}(n^3)$.

$M =$ "On input string $w = w_1 \cdots w_n$:

1. Let n be the length of w.
2. If n is not a perfect square, then *reject*.
3. Find integer m, such that $m^2 = n$.
4. Run TM for A on input $w_1 \cdots w_m$.
5. If TM for A rejects, then *reject*.
6. If some w_i in $w_{m+1} \cdots w_n$ is not symbol #, then *reject*.
7. *Accept*."

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